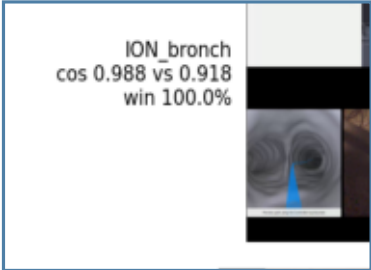


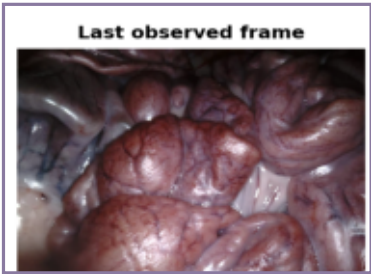
Laparoscopy



GI endoscopy

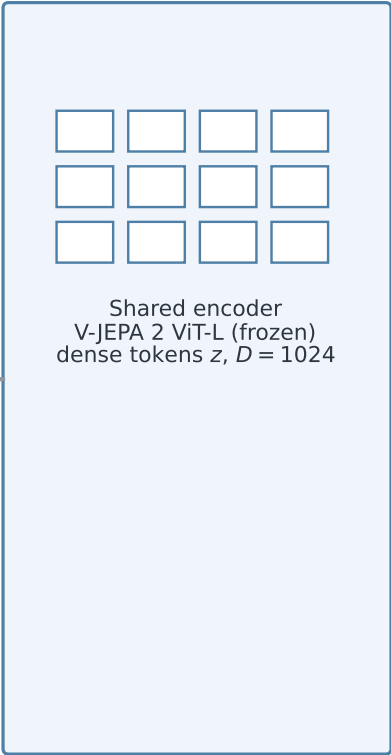


Bronchoscopy

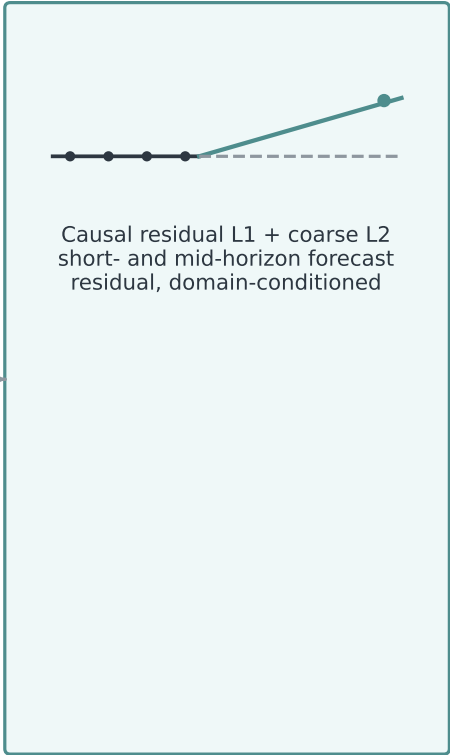


SCARED (pose + depth)

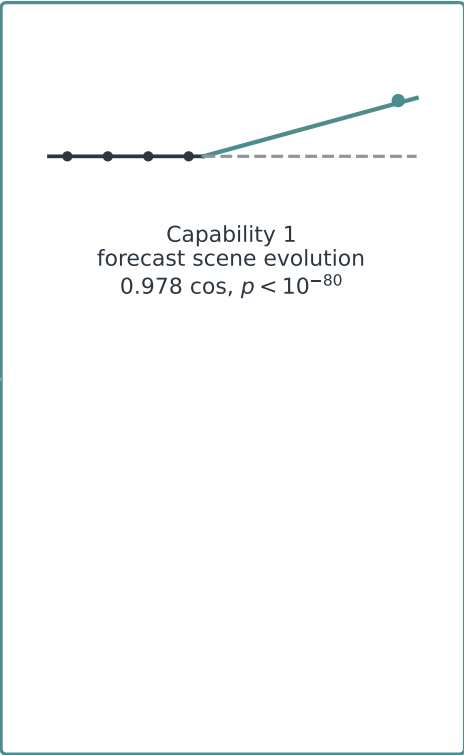
Passive-video forecast (validated, three orifices)



tokens



forecast



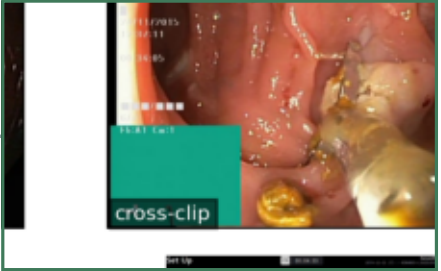
loss-of-view warning

camera-handling training

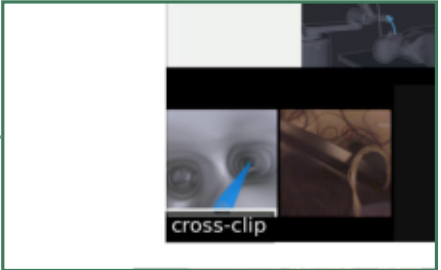
navigation assistance



cross-clip

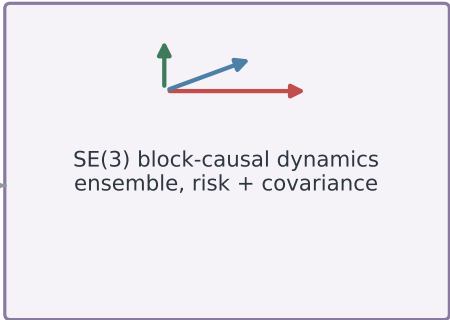


cross-clip

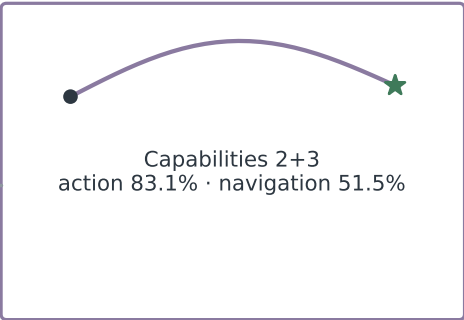


cross-clip

Physical grounding (audited; pose/depth-gated, SCARED/C3VD)



plan



SE(3)-conditioned retrieval

Action / navigation results (SCARED)

Input-sensitivity tests

history / action / domain

Reprojection pose gate

10.2 px -> 0.21 px

Matched negative banks

fixed, same-sequence

Grouped CV + frozen test

no test-set tuning